10.6: Spanning Trees and a Shortest Path Algorithm

Definition.
A **spanning tree** for a graph G is a subgraph of G that contains every vertex of G and is a tree.

Proposition 10.6.1

1. Every connected graph has a spanning tree.
2. Any two spanning trees for a graph have the same number of edges.

Definition.

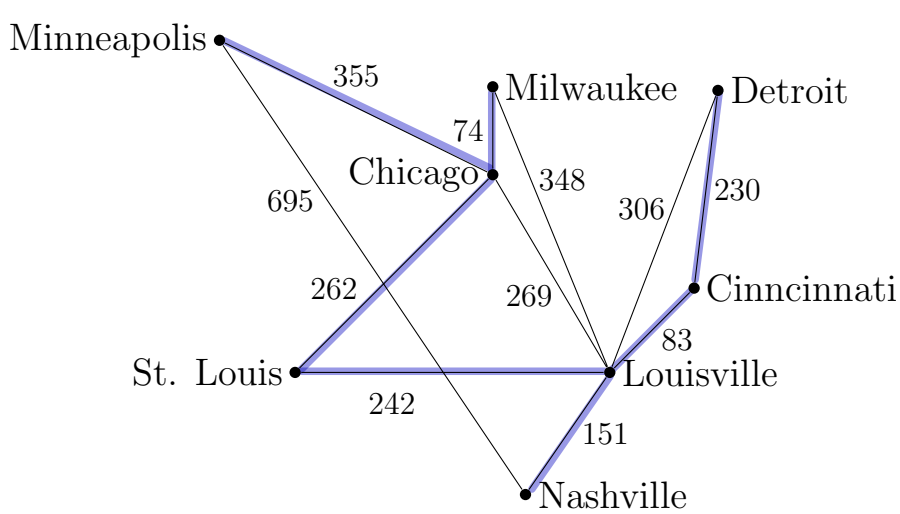
- A **weighted graph** is a graph for which each edge has an associated positive real number **weight**.
- The sum of the weights of all the edges is the **total weight** of the graph.
- A **minimum spanning tree** for a connected, weighted graph is a spanning tree that has the least possible total weight compared to all other spanning trees for the graph.

If G is a weighted graph and e is an edge of G, then $w(e)$ denotes the weight of e and $w(G)$ denotes the total weight of G.

Algorithm 3: Algorithm 10.6.1: Kruskal

Input : G [a connected, weighted graph with n vertices]
Initialize T to have all the vertices of G and no edges.
Let E be the set of all the edges of G, and let $m := 0$
while $m < n - 1$ **do**
  Find an edge e in E of least weight
  Delete e from E
  if *adding e to T does not produce a circuit* **then**
    add e to the edge set of T
    $m := m + 1$
end
Output: T [a minimum spanning tree for G]

Example. Use Kruskal's algorithm to find a minimum spanning tree of the following graph:



74 ✓
83 ✓
151 ✓
230 ✓
242 ✓
262 ✓
269 ✗ cycle
306 ✗ cycle
348 ✗ cycle
355 ✓
695 ✗ cycle

Algorithm 4: Algorithm 10.6.2: Prim's Algorithm

Input : G [a connected, weighted graph with n vertices]

Pick a vertex v of G and let T be the graph with one vertex, v, and no edges

Let V be the set of all the vertices of G except v

for $i := 1$ ***to*** $n - 1$ **do**

Find an edge e of G such that

1. e connects T to one of the vertices in V, and
2. e has the least weight of all edges connecting T to a vertex in V

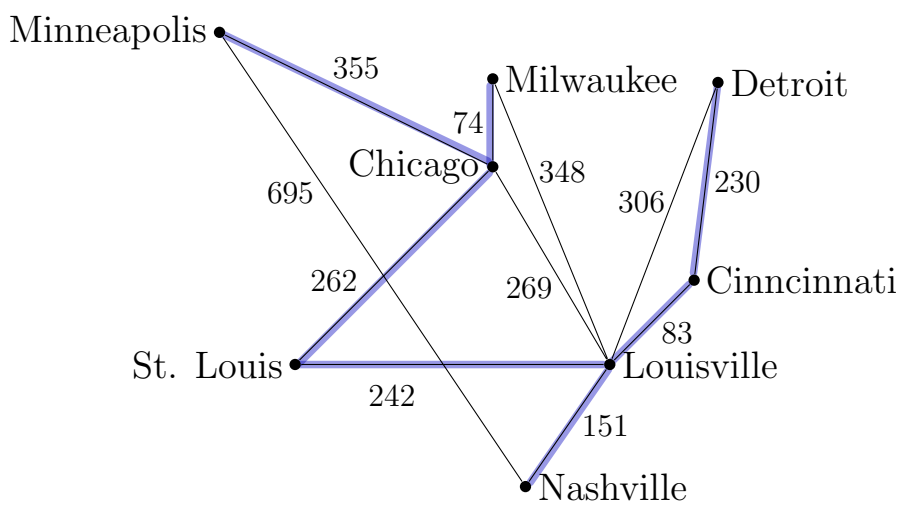
Let w be the endpoint of e that is in V.

Add e and w to the edge and vertex sets of T, and delete w from V

end

Output: T [a minimum spanning tree for G]

Example. Use Prim's algorithm to find a minimum spanning tree of the following graph:



Candidate edges that don't introduce cycles ↓

74

262 $= \min\{355, 262, 269, 348\}$

242 $= \min\{355, \cancel{269}, 348, 242\}$ ↑cycle

83 $= \min\{355, 348, 83, 151\}$

151 $= \min\{355, 348, 151, 230\}$

230 $= \min\{355, 348, 230, 695\}$

355 $= \min\{355, 348, 695\}$